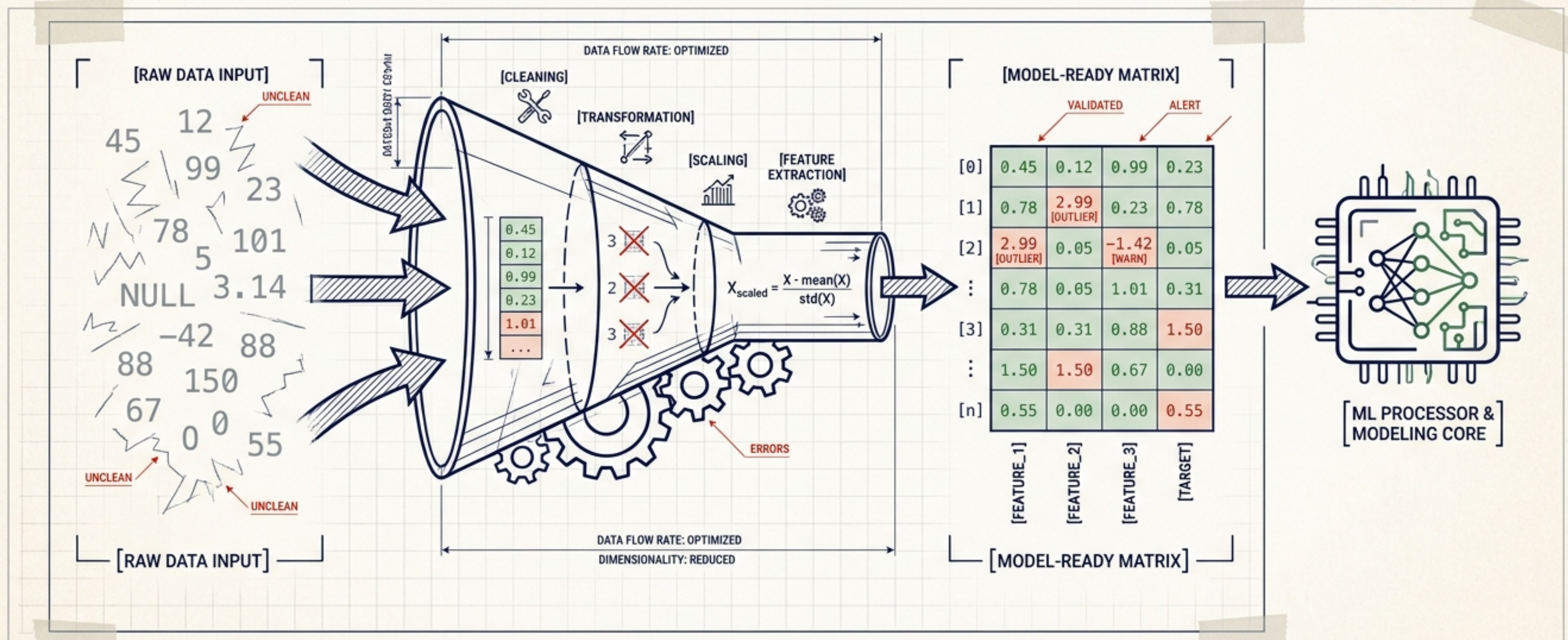


Statistics for Machine Learning

From Raw Data to Model-Ready Intelligence

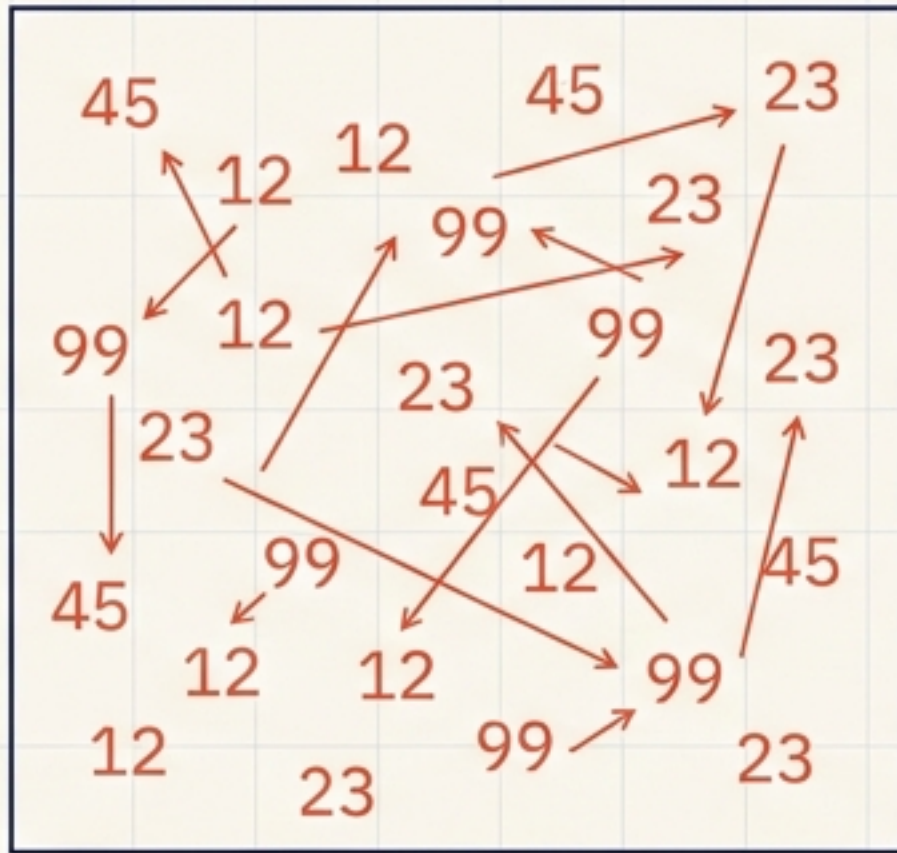
The Blueprint for Students & Practitioners



The Core Mission: Making Data Speak

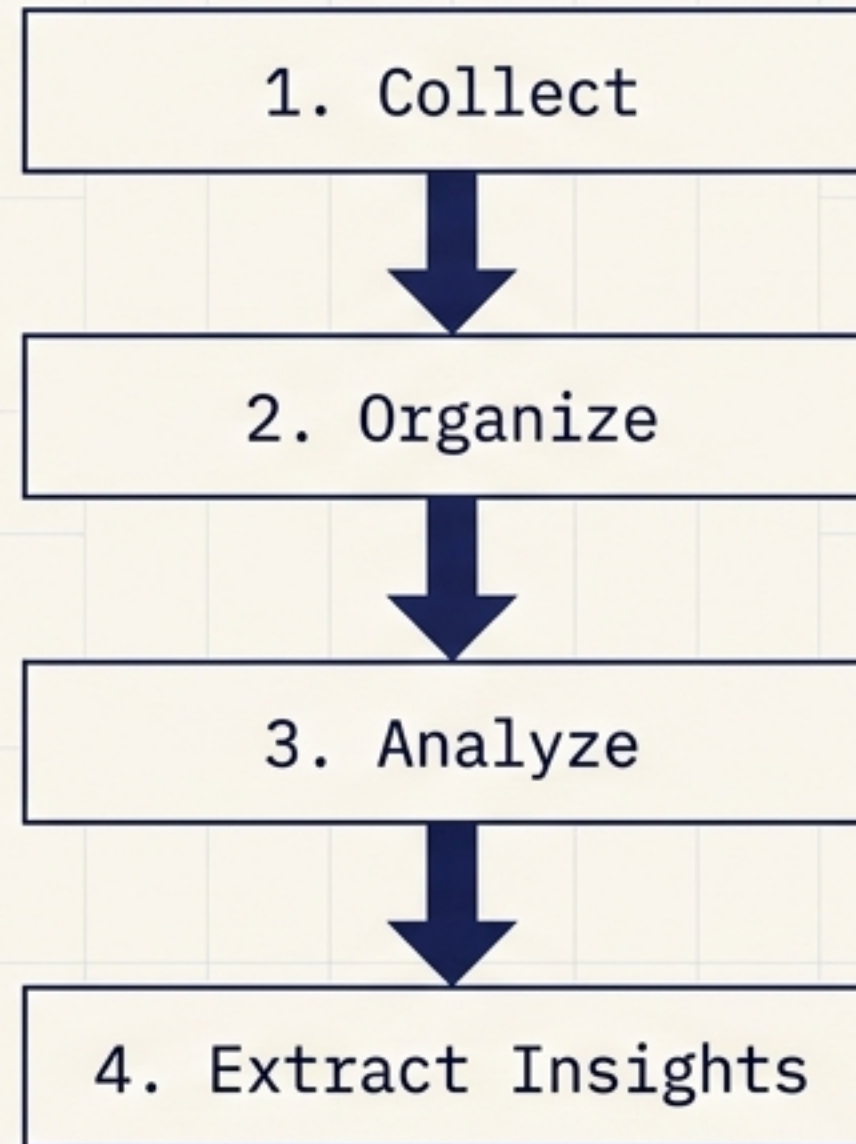
Statistics is the mathematical engine that transforms meaningless noise into structured, actionable insights.

The Noise



Scattered Data: Meaningless noise. No insights can be extracted.

The 4-Step Engine



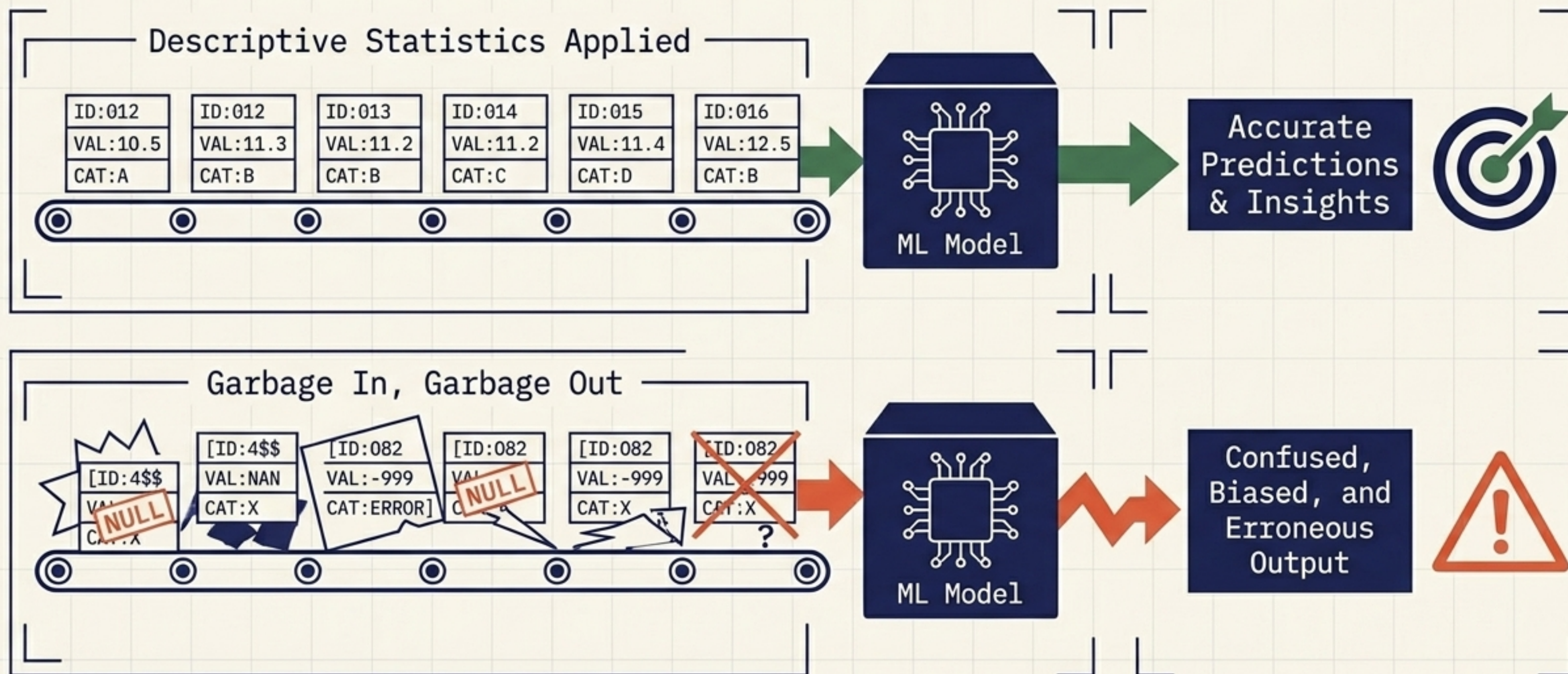
The Signal

StdId	Roll	Marks
001	12	45
002	23	99
003	45	12
004	99	23

Structured Data: Understandable, quantifiable, and actionable.

The Golden Rule of ML: Garbage In, Garbage Out

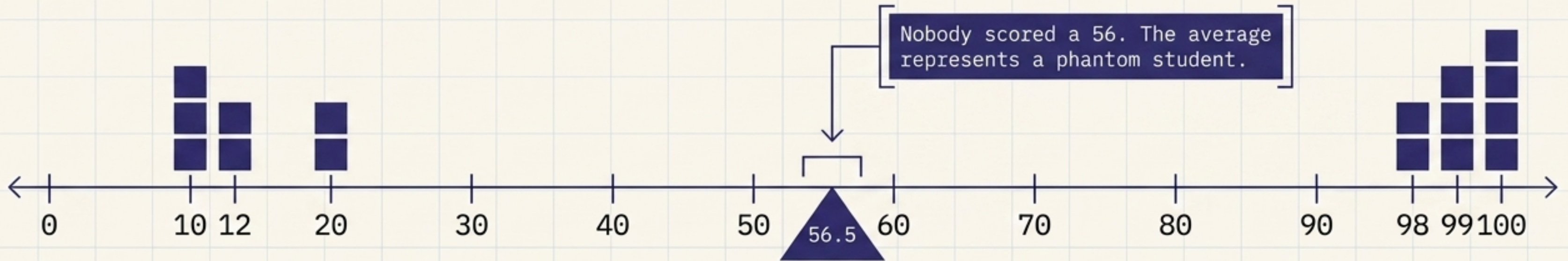
Machine Learning models are entirely dependent on data quality. Descriptive statistics measures, understands, and fixes the 'garbage' before the model ever sees it.



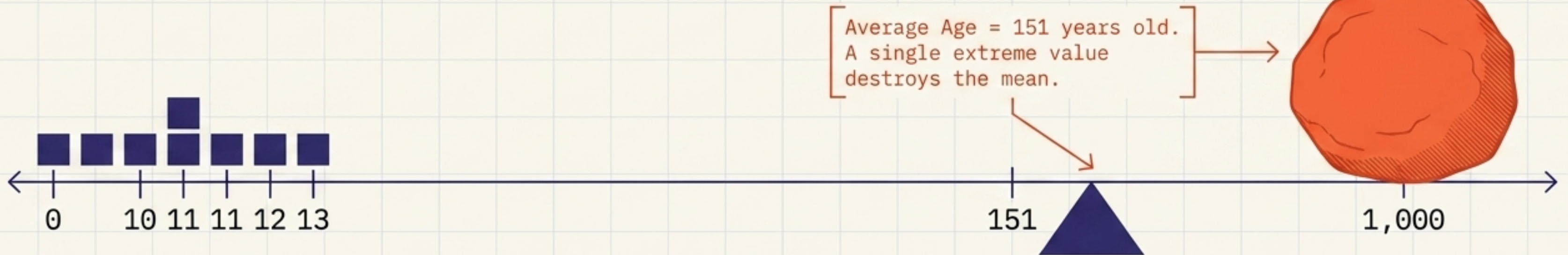
The Problem with 'Average'

We use Mean, Median, and Mode to find the center. But when extreme values (Outliers) exist, the Average lies.

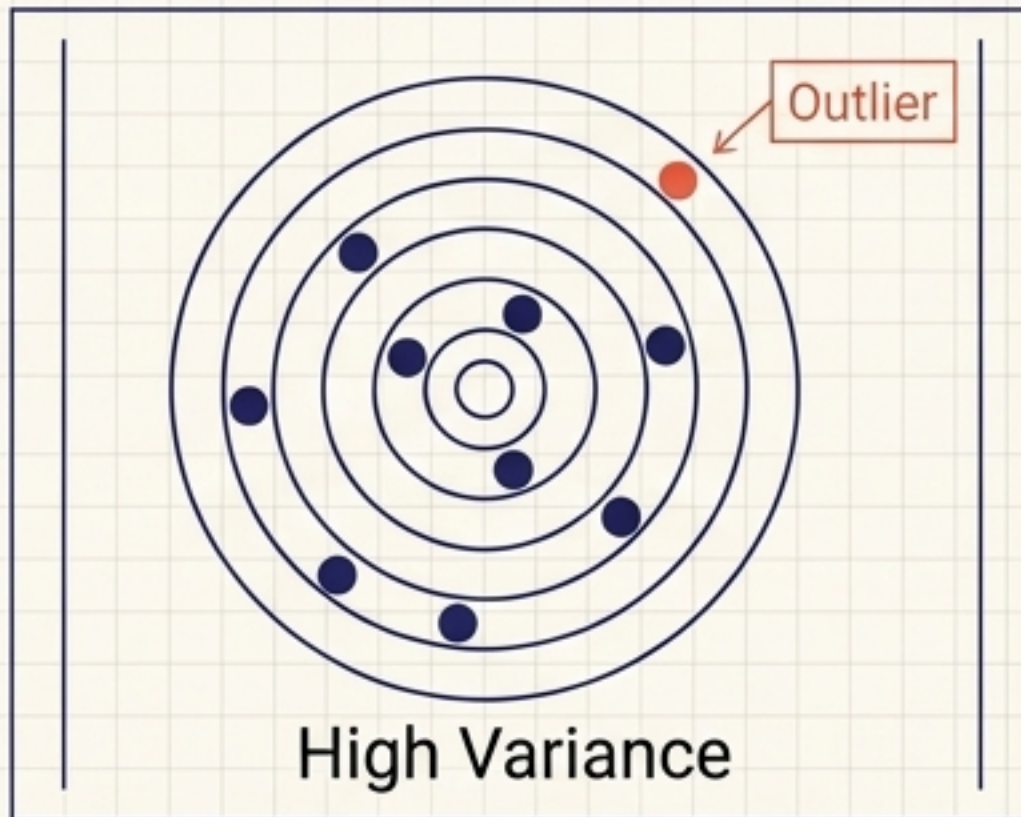
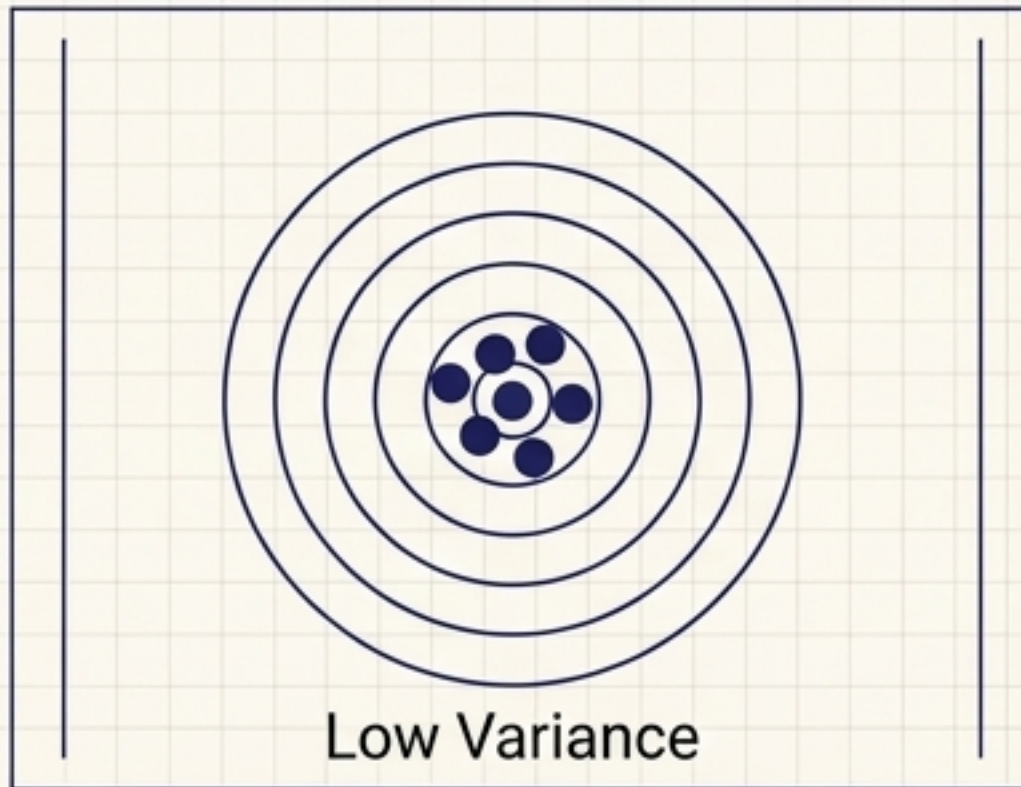
The Bimodal Trap



The Outlier Trap



Understanding Spread: Variance (σ^2)



Tracking the Outlier Dataset (Ages: 0, 10, 11, 11, 12, 13, 1000)

Step 1: The Mean

$$\text{Sum (1057)} / N (7) = \mu \text{ 151}$$

Step 2: The Distance

$$(1000 - 151)^2 = 720,801 \text{ (The outlier explodes the calculation)}$$

Step 3: The Variance (σ^2)

$$\text{Sum of squares (841,048)} / 7 = 120,149.71$$

The ML Impact: High variance (120,149) screams 'Noise!' The data is not uniformly distributed. An ML model fed this data will learn the wrong patterns.

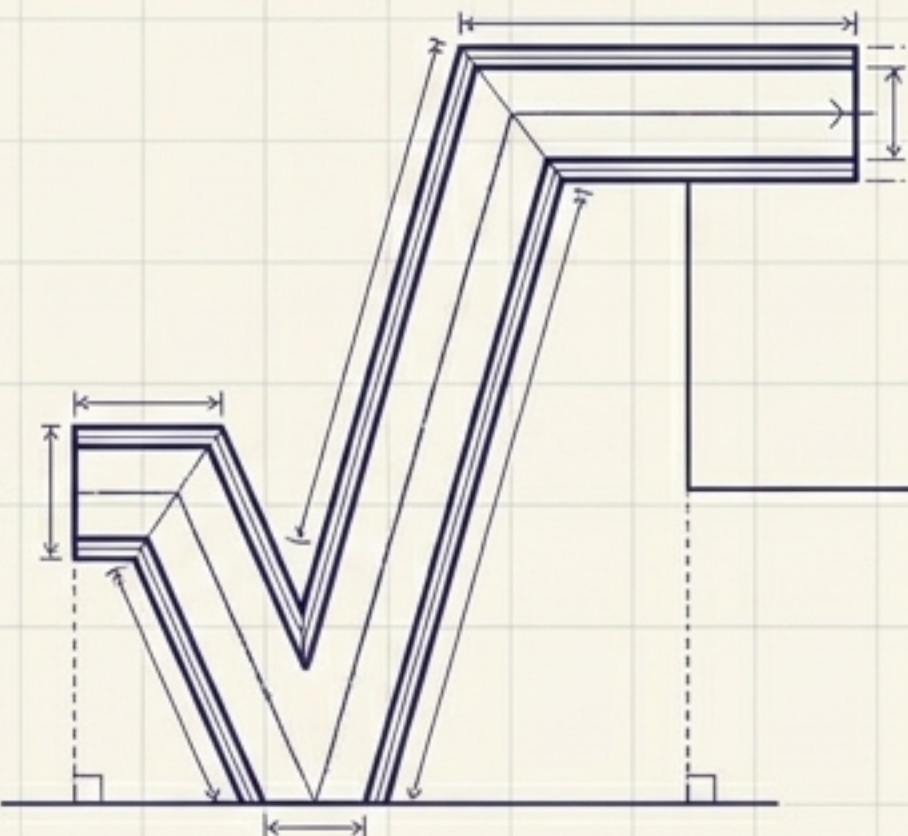
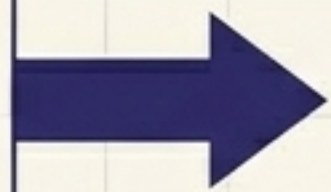
The Solution: Standard Deviation (σ)

Variance squares the units, making it impossible to interpret. We apply a square root to return to the original unit.

Transformation Filter

120,149

(Age²)



346.62

(Years)

The Reality Check: An SD of 346 means the ages in our dataset deviate from the mean by an average of 346 years.

Low Standard Deviation:
Tightly clustered, clean data (Good for ML).

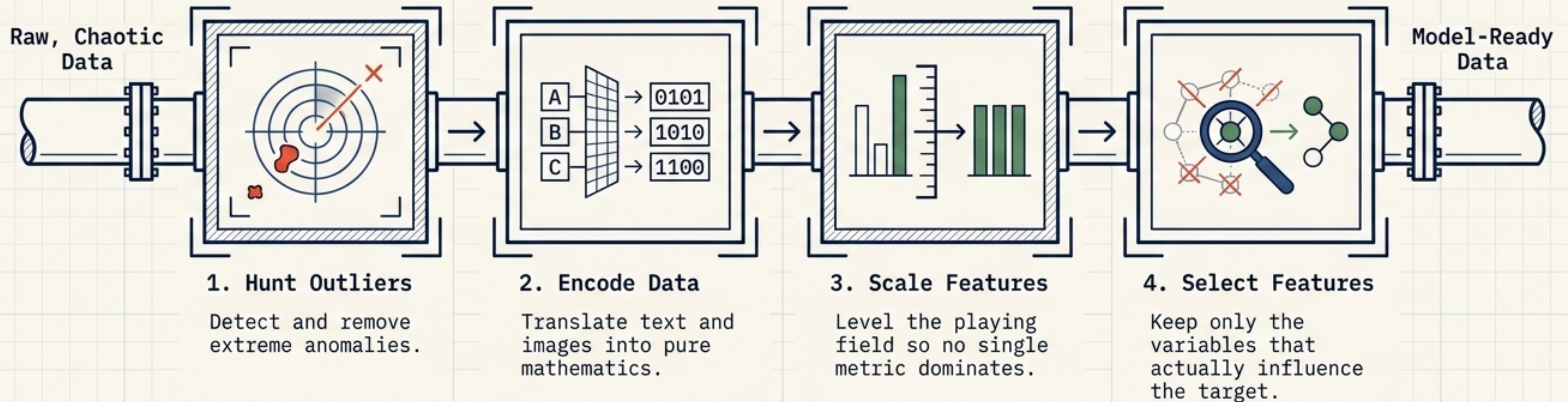
High Standard Deviation:
Scattered, noisy data with outliers (Bad for ML).

Comparison Matrix: Variance vs. Standard Deviation

Attribute	Variance	Standard Deviation
Mathematical Relationship	σ^2 (Square)	σ (Root)
Unit of Measurement	Squared (e.g., TK ² , Age ²)	Original (e.g., TK, Age)
Scale & Size	Unnaturally Large	Realistic and Interpretable
Human Interpretability	Difficult to visualize directly	Easy to understand (average distance from mean)
Machine Learning Use	Used internally in Cost Functions	Used for Feature Scaling (Standardization/Z-Score)

The Data Preprocessing Pipeline

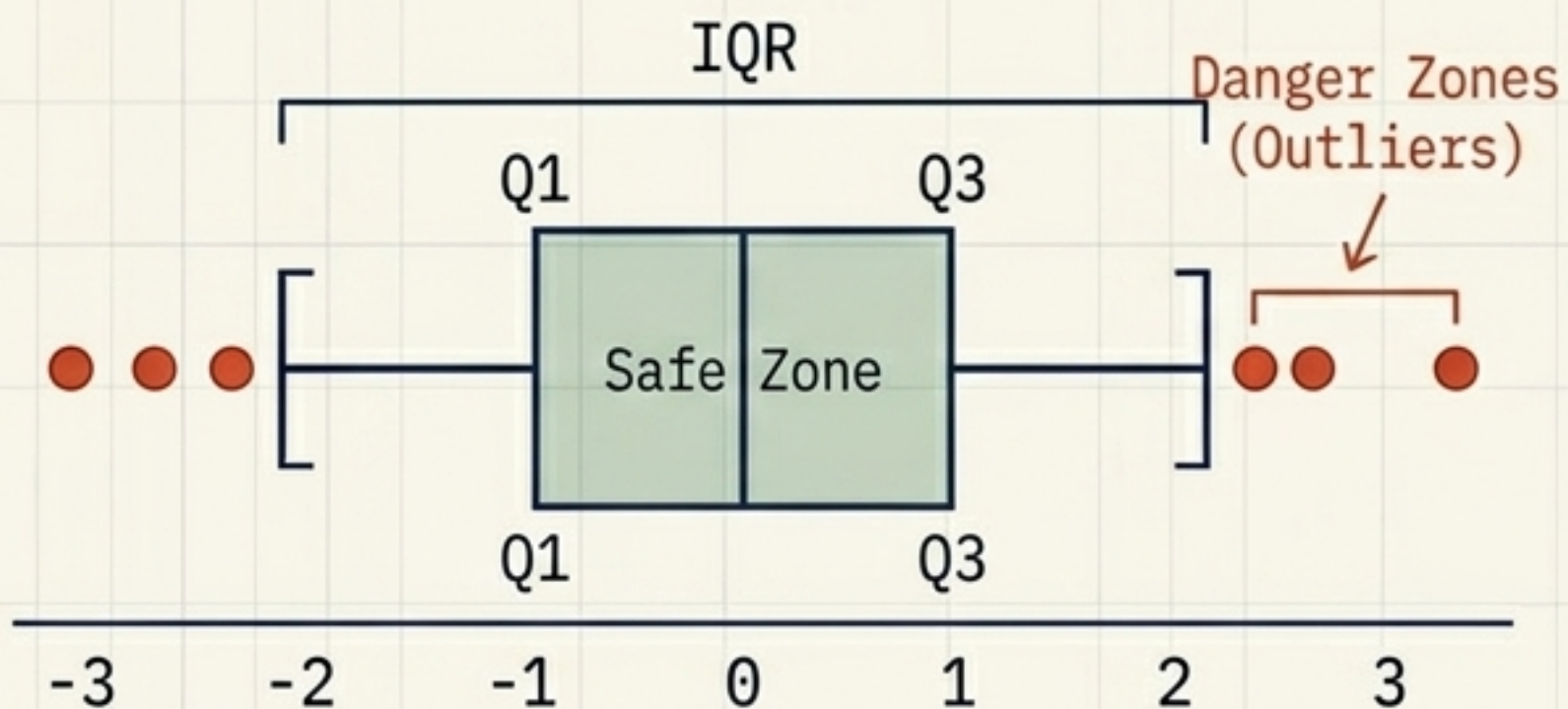
Raw data is fundamentally incompatible with Machine Learning. Preprocessing is the statistical bridge.



Pillar 1: Hunting Outliers

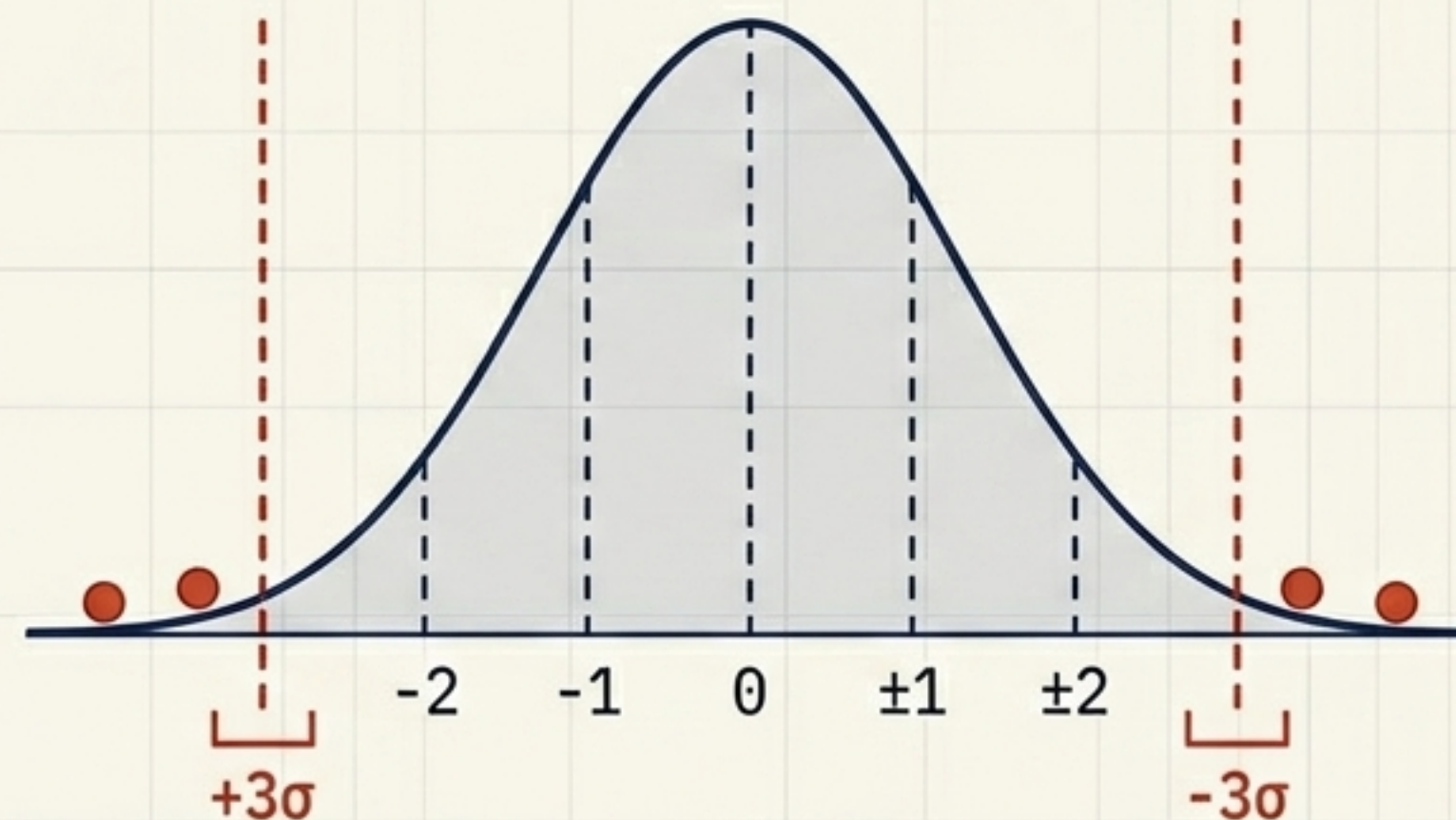
Outliers bias models. We trap them by setting rigid mathematical boundaries based on data density or deviation.

IQR (Interquartile Range)



Sets mathematical boundaries based on the middle 50% of data.

Z-Score

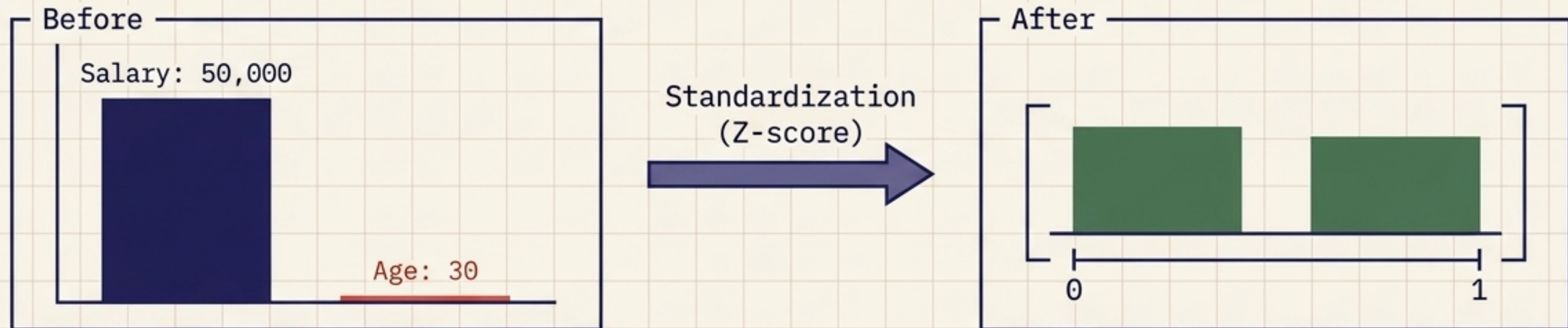


Flags any data point that strays more than 3 Standard Deviations from the norm.

Pillars 2 & 3: Encoding and Scaling



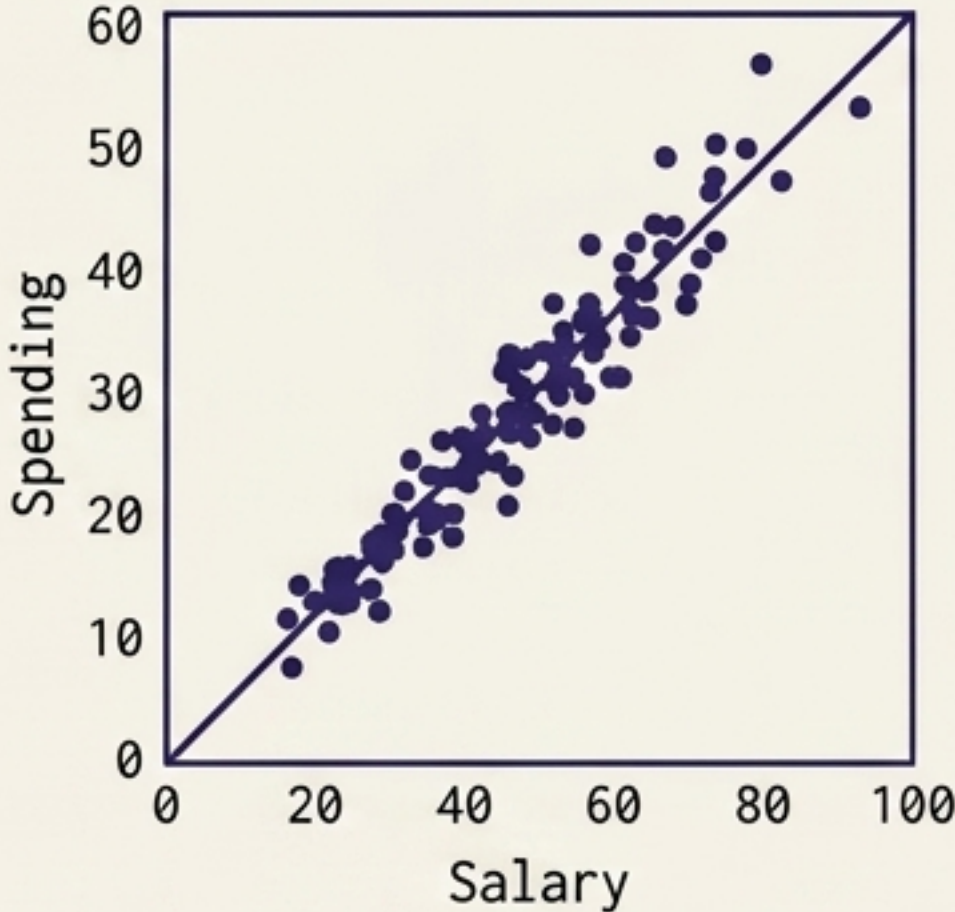
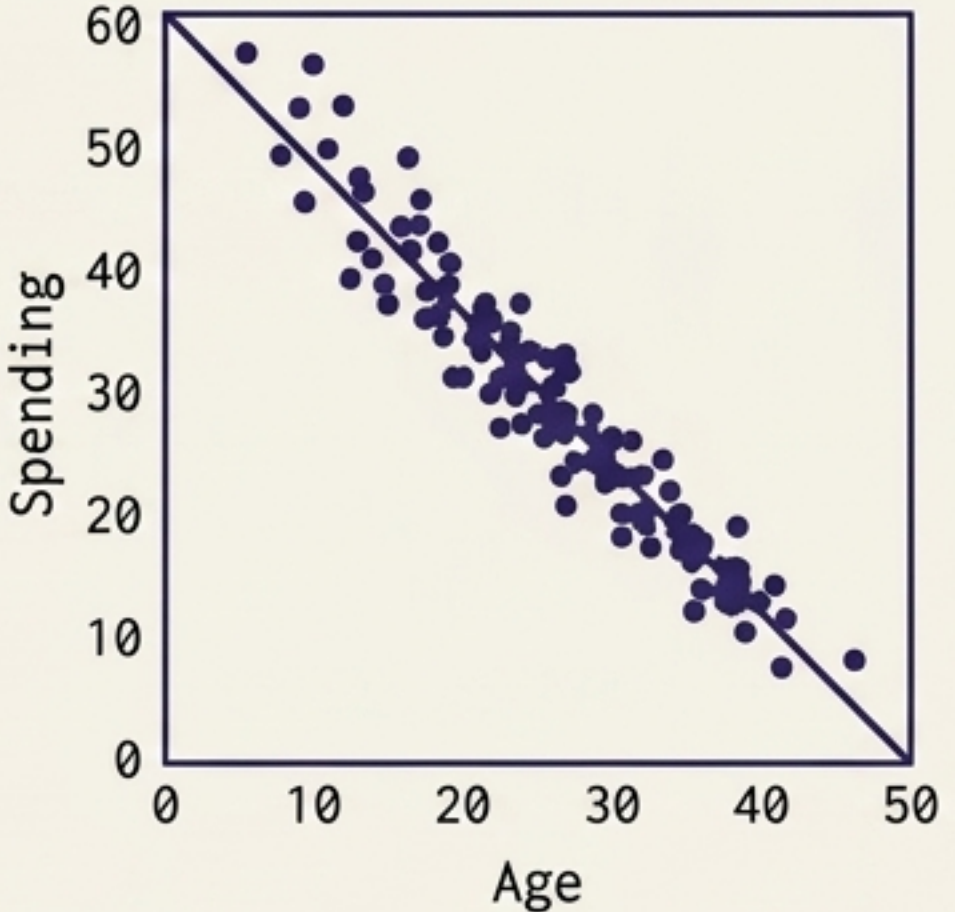
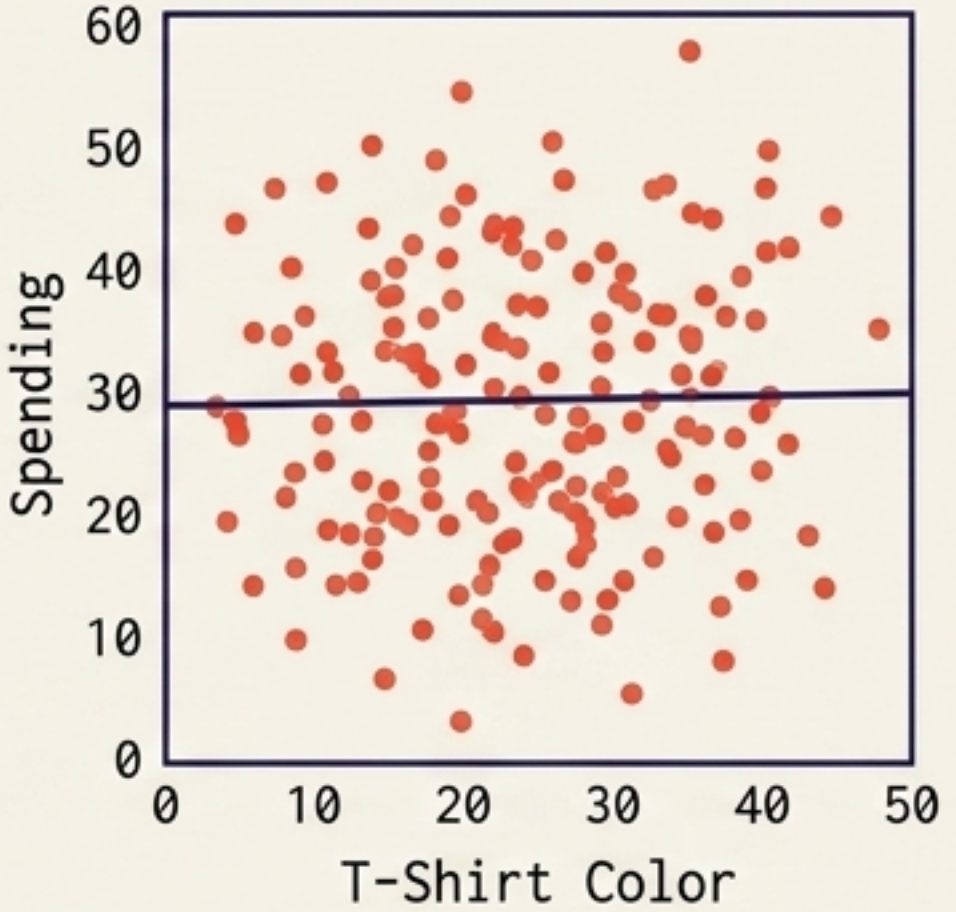
Encoding: ML only speaks Math. Text and Pixels must become numbers.



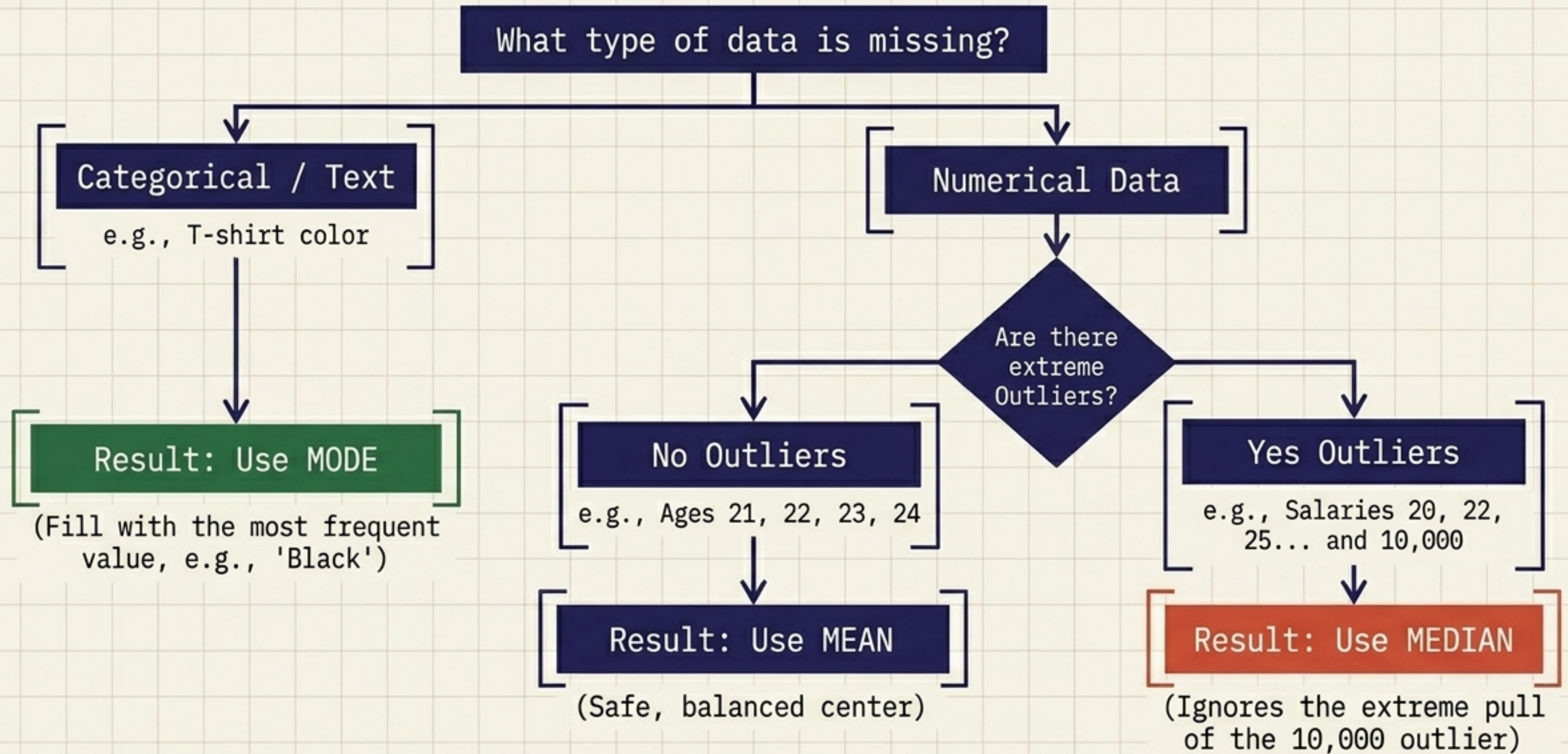
Feature Scaling: Without scaling, a model assumes a 50,000 Salary is infinitely more important than a 30-year Age. Scaling forces equal mathematical representation.

Pillar 4: Feature Selection (Correlation)

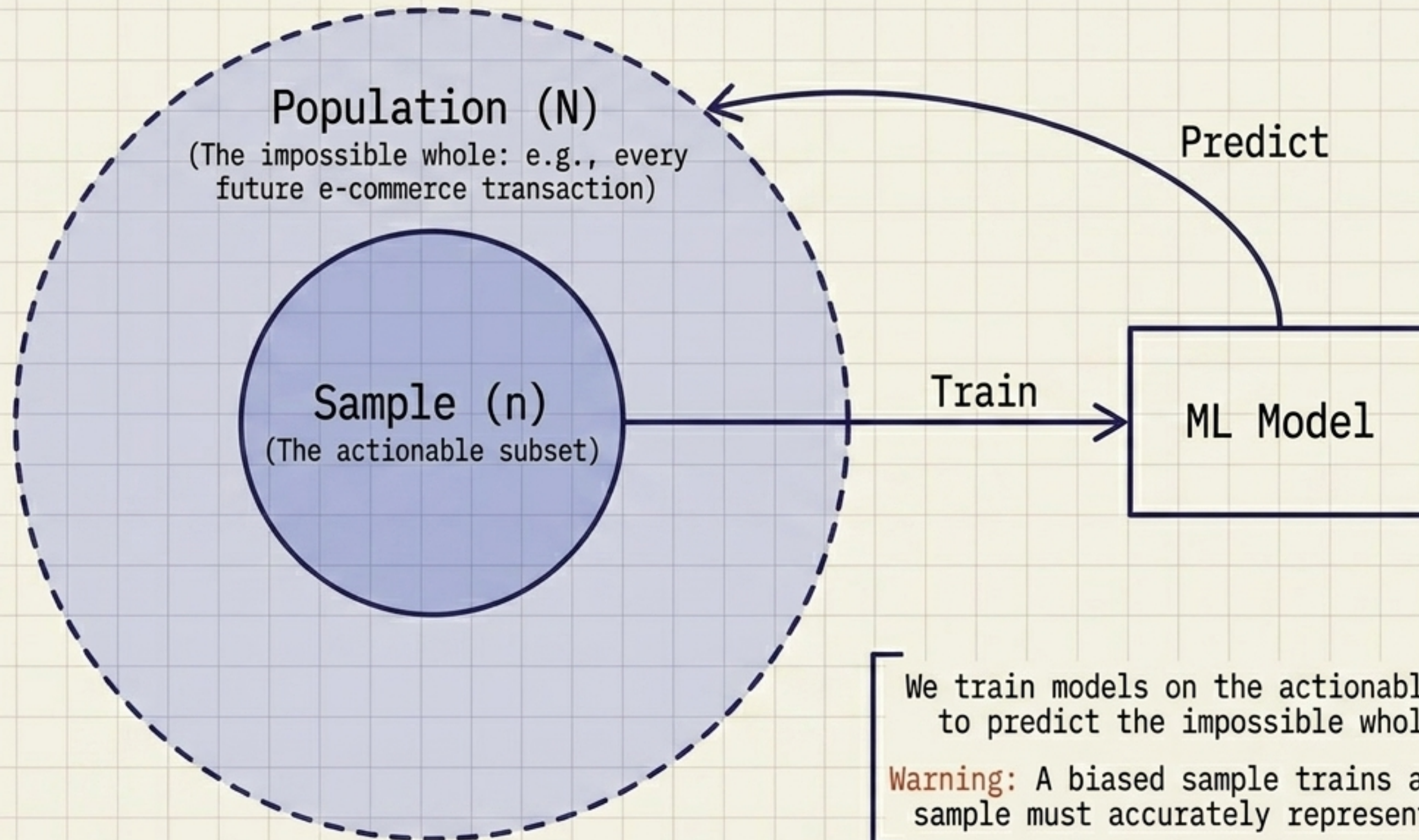
Don't bloat the model with useless data. Use Correlation to find relationships to the Target.

Positive Correlation	Negative Correlation	Zero Correlation
		
Highly valuable pattern.	Inverse relationship. Highly valuable.	Dead weight. Drop it.

The Missing Value Decision Tree



Inferential Statistics: The Big Picture



The Data Pre-Flight Checklist

- [✓] **Outliers Removed?** (If found, Mean is broken. Use Median).
- [✓] **Data Spread Checked?** (Low SD = Good. High SD = Needs cleaning).
- [✓] **Missing Values Handled?** (Mean for normal, Median for skewed, Mode for text).
- [✓] **Features Scaled?** (Ensure Age and Salary have an equal mathematical voice).
- [✓] **Relevant Features Selected?** (Drop zero-correlation noise like T-Shirt color).



Final Output: Clean Data → High-Performing Machine Learning Model